

Exam — Problems & Solutions

April 11, 2026

Problem 1

Problem. Given an LTI system, the input signal is

$$x[n] = u[n] - u[n - 10],$$

and the impulse response is

$$h[n] = \left(\frac{1}{3}\right)^n u[n].$$

Calculate the value of the output signal $y[n]$ when $-3 \leq n \leq 3$.

Solution

Since

$$x[n] = u[n] - u[n - 10] = \begin{cases} 1, & 0 \leq n \leq 9, \\ 0, & \text{otherwise,} \end{cases}$$

we have

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = \sum_{k=0}^9 \left(\frac{1}{3}\right)^{n-k} u[n - k].$$

For $n < 0$, we have $u[n - k] = 0$ for all $k \geq 0$, so

$$\boxed{y[n] = 0, \quad n < 0.}$$

For $0 \leq n \leq 9$, only the terms $0 \leq k \leq n$ remain, hence

$$y[n] = \sum_{k=0}^n \left(\frac{1}{3}\right)^{n-k} = \sum_{m=0}^n \left(\frac{1}{3}\right)^m = \frac{1 - (1/3)^{n+1}}{1 - 1/3}.$$

Therefore

$$\boxed{y[n] = \frac{3}{2} \left(1 - \left(\frac{1}{3}\right)^{n+1}\right), \quad 0 \leq n \leq 9.}$$

Since the question only asks for $-3 \leq n \leq 3$, these are the only two relevant cases. Thus

n	-3	-2	-1	0	1	2	3
$y[n]$	0	0	0	1	$\frac{4}{3}$	$\frac{13}{9}$	$\frac{40}{27}$

Problem 2

Problem. An LTI system is described by the input-output relation

$$y[n] = x[n] + 2x[n-1] + x[n-2].$$

(a) Determine $h[n]$, the impulse response of the system, and $H(e^{j\omega})$, the frequency response of the system. (b) Is this a stable system?

Solution

(a) For an LTI system,

$$y[n] = x[n] * h[n].$$

Comparing with

$$y[n] = x[n] + 2x[n-1] + x[n-2],$$

we directly obtain

$$h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2].$$

Taking the DTFT gives

$$H(e^{j\omega}) = 1 + 2e^{-j\omega} + e^{-j2\omega}.$$

An equivalent factored form is

$$H(e^{j\omega}) = (1 + e^{-j\omega})^2 = 4e^{-j\omega} \cos^2\left(\frac{\omega}{2}\right).$$

(b) The system is stable because $h[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| = 1 + 2 + 1 = 4 < \infty.$$

Hence the system is BIBO stable.

$$\text{The system is stable.}$$

Problem 3

Problem. Consider ideal impulse-train sampling followed by reconstruction. The input signal is

$$x_c(t) = 2\cos(20\pi t) + \cos(50\pi t), \quad -\infty < t < \infty.$$

The reconstruction filter has frequency response

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| \leq \pi/T, \\ 0, & |\Omega| > \pi/T. \end{cases}$$

(a) Determine the continuous-time Fourier transform $X_c(j\Omega)$ and sketch it as a function of Ω . (b) Assume that $f_s = 1/T = 60$ samples/sec. Plot $X_s(j\Omega)$ for $-2\pi/T \leq \Omega \leq 2\pi/T$. What is the output $x_r(t)$?

Solution

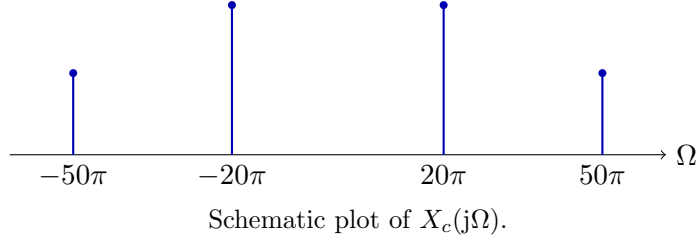
(a) Using the CTFT pair

$$A \cos(\Omega_0 t) \longleftrightarrow A\pi [\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)],$$

we obtain

$$X_c(j\Omega) = 2\pi\delta(\Omega - 20\pi) + 2\pi\delta(\Omega + 20\pi) + \pi\delta(\Omega - 50\pi) + \pi\delta(\Omega + 50\pi).$$

So $X_c(j\Omega)$ consists of impulses at $\Omega = \pm 20\pi$ and $\Omega = \pm 50\pi$.



Sampling by the impulse train $s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ gives

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)), \quad \Omega_s = \frac{2\pi}{T}.$$

Thus $X_s(j\Omega)$ is a train of shifted replicas of $X_c(j\Omega)$, spaced by Ω_s .

(b) $f_s = 60$ **samples/sec.** Here

$$T = \frac{1}{60}, \quad \Omega_s = \frac{2\pi}{T} = 120\pi, \quad \frac{\pi}{T} = 60\pi.$$

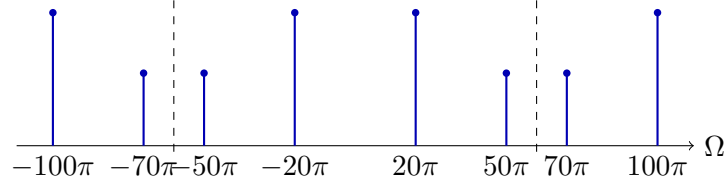
Hence

$$X_s(j\Omega) = 60 \sum_{k=-\infty}^{\infty} X_c(j(\Omega - 120\pi k)).$$

Over the interval $-2\pi/T \leq \Omega \leq 2\pi/T$, i.e. $-120\pi \leq \Omega \leq 120\pi$, the nonzero impulses are

Ω	coefficient in $X_s(j\Omega)$
-100π	120π
-70π	60π
-50π	60π
-20π	120π
20π	120π
50π	60π
70π	60π
100π	120π

The baseband replica lies entirely inside the reconstruction-filter passband $|\Omega| \leq 60\pi$, so there is no aliasing.



Schematic plot of $X_s(j\Omega)$ for $f_s = 60$ Hz. The dashed lines mark the passband $|\Omega| \leq 60\pi$.

After reconstruction,

$$X_r(j\Omega) = H_r(j\Omega)X_s(j\Omega) = \begin{cases} TX_s(j\Omega), & |\Omega| \leq 60\pi, \\ 0, & |\Omega| > 60\pi. \end{cases}$$

Only the $k = 0$ replica survives. Therefore

$$X_r(j\Omega) = X_c(j\Omega),$$

and hence

$$x_r(t) = 2 \cos(20\pi t) + \cos(50\pi t).$$

So the signal is reconstructed exactly.

Problem 4

Problem. Consider a LTI system defined as

$$H(z) = \frac{1}{1 - 1.6z^{-1} + 0.65z^{-2}}.$$

(a) State the region of convergence for $H(z)$ to be causal. (b) If the system is causal, determine whether it is also stable.

Solution

First find the poles from

$$z^2 - 1.6z + 0.65 = 0.$$

Thus

$$z = \frac{1.6 \pm \sqrt{1.6^2 - 4(0.65)}}{2} = \frac{1.6 \pm j0.2}{2} = 0.8 \pm 0.1j.$$

So both poles have magnitude

$$|0.8 \pm 0.1j| = \sqrt{0.8^2 + 0.1^2} = \sqrt{0.65}.$$

(a) For a causal rational system, the ROC is outside the outermost pole. Hence

$$\text{ROC}(H) : |z| > \sqrt{0.65}.$$

(b) Yes, the causal system is stable. Since

$$\sqrt{0.65} < 1,$$

both poles lie inside the unit circle. Equivalently, for the causal ROC $|z| > \sqrt{0.65}$, the unit circle $|z| = 1$ is included in the ROC. Therefore the system is BIBO stable.

$$\text{If the system is causal, then it is also stable.}$$